

Statistical Theory and Modeling (ST2601)

Lecture 6 - The central theorems, transformations and Monte Carlo

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Overview

- Stochastic convergence
- Law of large numbers
- Central limit theorem
- Transformations of random variables
- Monte Carlo simulation

Stochastic convergence - asymptotics

- Performance of a statistical method in large samples $n \rightarrow \infty$.
- Can be a good approximation for finite samples.
- **Sequence of random variables** $X_1, X_2, \dots, X_n$.
- Example: sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- What happens with X_n as $n \rightarrow \infty$?
 - ▶ Does it concentrate on a single value?
 - ▶ Does the distribution of X_n stabilize?

Convergence in distribution

- The sequence $X_1, X_2, \dots, X_n$ **converges in distribution** to the random variable X if “**the cdf of X_n starts to look like the cdf of X** ” when n gets large.
- $F_n(x)$ is the cdf of X_n
- $F(x)$ is the cdf of X

Definition. A sequence of random variables $X_1, \dots, X_n$ **converges in distribution** to the random variable X , if

$$F_n(x) \rightarrow F(x) \quad \text{as } n \rightarrow \infty,$$

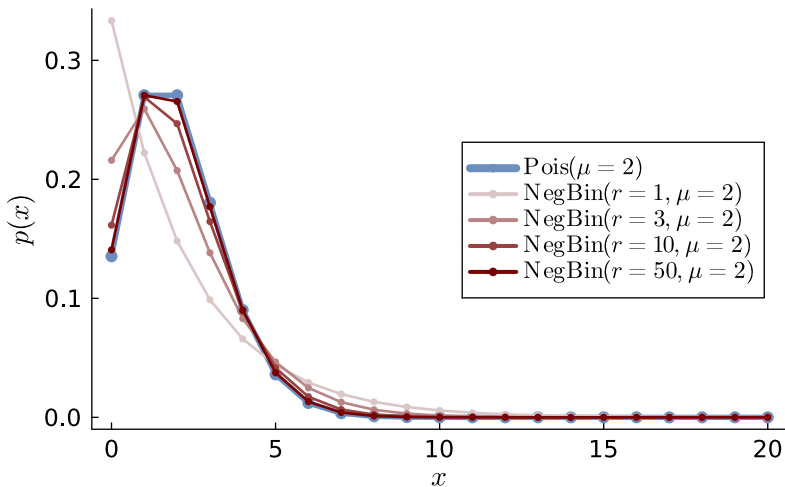
for all x where $F(\cdot)$ is continuous, where $F_n(x)$ and $F(x)$ are the cumulative distribution functions (cdf) of X_n and X , respectively.

We then write $X_n \xrightarrow{d} X$.

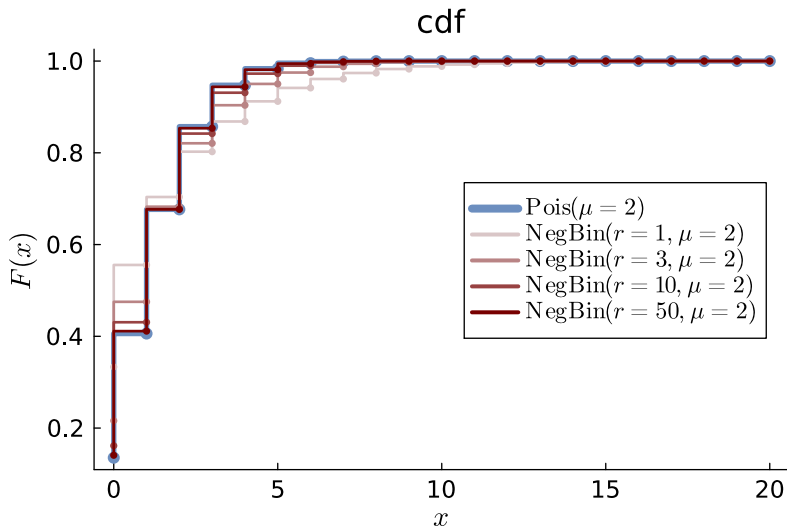
NegBin converges in distribution to Poisson

$$\text{NegBin}(r, \mu) \xrightarrow{d} \text{Pois}(\mu) \text{ as } r \rightarrow \infty$$

pdf

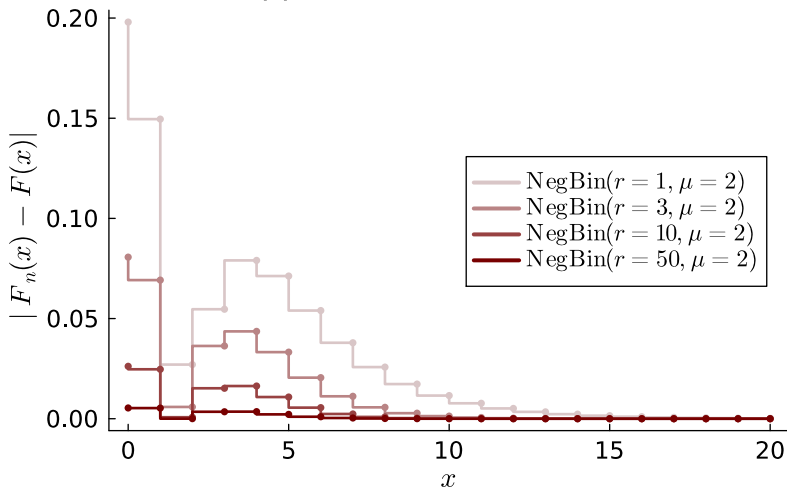


NegBin converges in distribution to Poisson

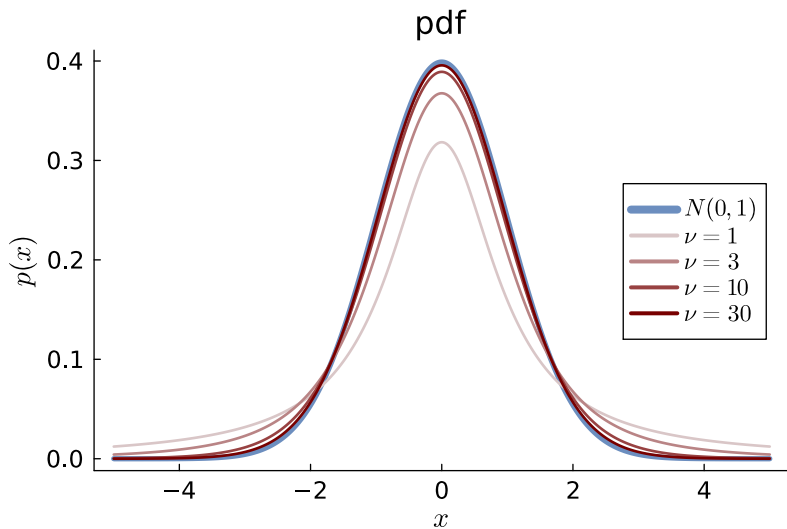


NegBin converges in distribution to Poisson

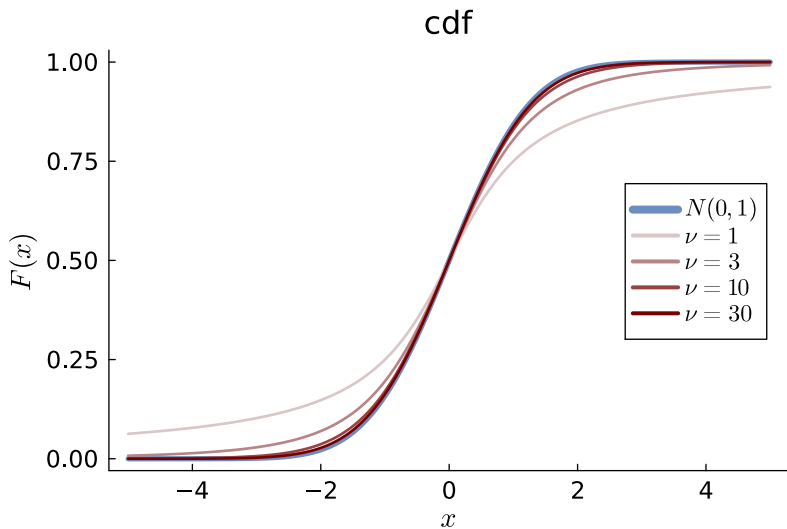
approximation error in cdf



Student- t converges in distribution to $N(0, 1)$

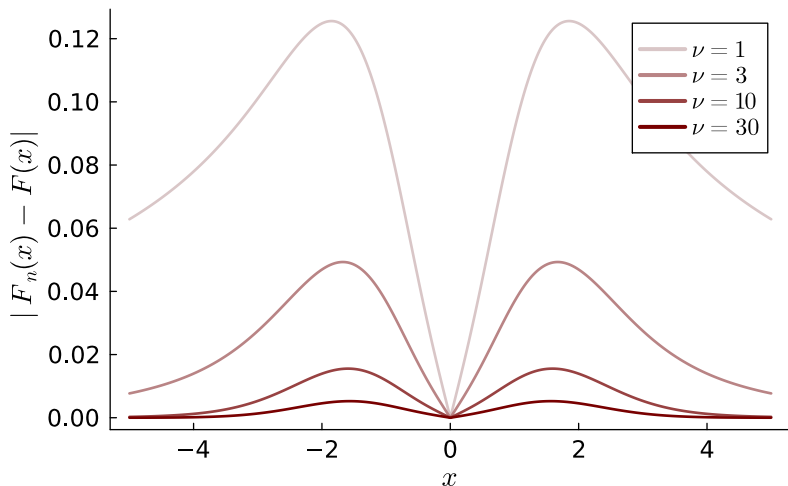


Student- t converges in distribution to $N(0, 1)$



Student- t converges in distribution to $N(0, 1)$

approximation error in cdf



Limit of a deterministic sequence

- **Mathematical limit** at infinity for **deterministic sequences**

$$\lim_{n \rightarrow \infty} x_n = L$$

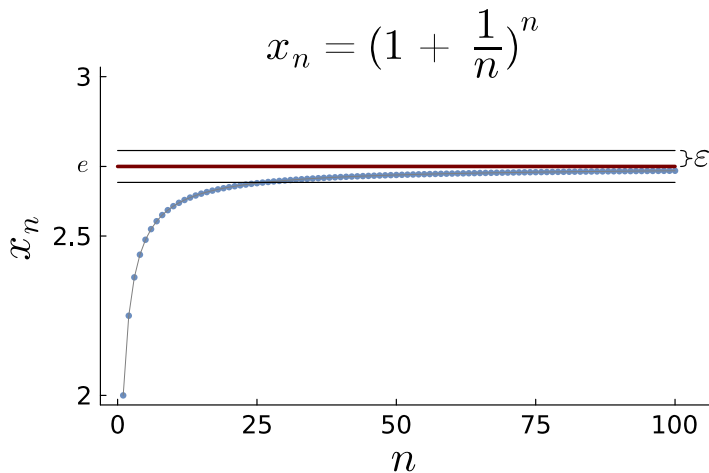
means that we can make sure that

$$|x_n - L| < \epsilon \quad \Longleftrightarrow \quad x_n \in (L - \epsilon, L + \epsilon)$$

for any $\epsilon > 0$, by choosing **a large enough n** .

- Example: $x_n = \left(1 + \frac{1}{n}\right)^n$, with $\lim_{n \rightarrow \infty} x_n = e \approx 2.7183$.
- X_n are random variables, cannot guarantee that $|X_n - L| < \epsilon$.

Limit of a deterministic sequence



Convergence in probability

- The sequence $X_1, X_2, \dots, X_n$ **converges in probability** to the constant c if “**the distribution of X_n concentrates around c** ” when n gets large.

Definition. A sequence of random variables $X_1, \dots, X_n$ *converges in probability to a constant c , if for all $\epsilon > 0$*

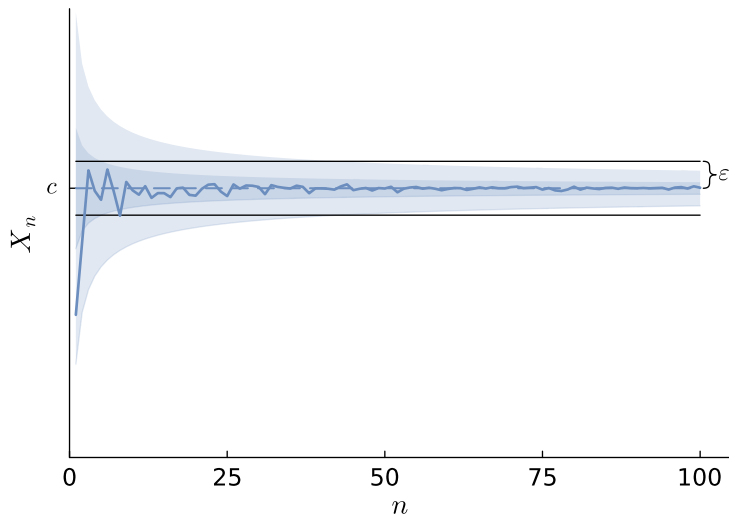
$$\Pr(|X_n - c| > \epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

We then write $X_n \xrightarrow{p} c$.

- We can also have convergence in probability to a random variable X instead of a constant; see the prequel book.

Convergence in probability

■ 50% and 95% probability intervals.



Law of large numbers

- The **law of large numbers** tells us that the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges in probability to the population mean $\mu = \mathbb{E}(X_i)$ as $n \rightarrow \infty$.

Theorem 4 (law of large numbers).

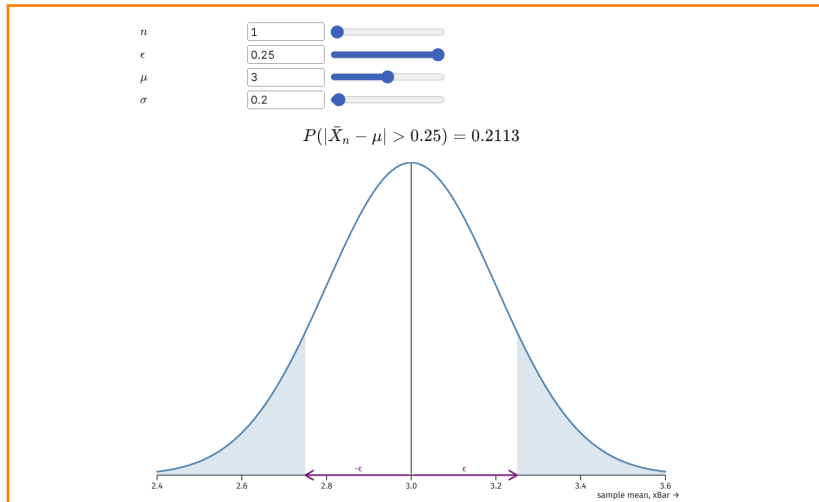
For independent random variables $X_1, X_2, \dots$ with finite mean $\mu = \mathbb{E}(X)$ and finite variance we have

$$\bar{X}_n \xrightarrow{p} \mu$$

where $\xrightarrow{p}$ denotes convergence in probability, i.e., for all $\epsilon > 0$

$$\Pr(|\bar{X}_n - \mu| \geq \epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad (5.3)$$

Law of large numbers - widget



Central limit theorem

- The **central limit theorem** tells us that the sample mean $\bar{X}_n$ **converges in distribution** to a normal distribution.

Theorem 6 (central limit theorem - informal version).

Let $X_1, X_2, \dots$ be iid random variables with finite mean μ and variance σ^2 . Then for large n ,

$$\bar{X}_n \overset{\text{approx}}{\sim} N(\mu, \sigma^2/n)$$

- Have to **standardize** to avoid a degenerate distribution:

$$\frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} = \sqrt{n} \frac{(\bar{X}_n - \mu)}{\sigma}$$

- Formal version

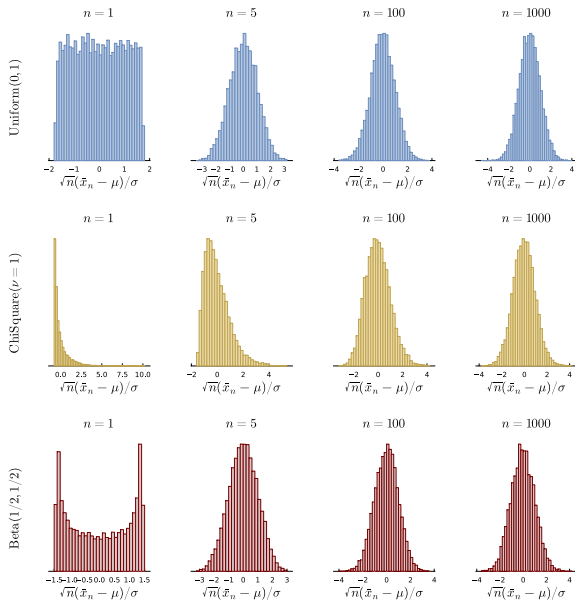
Theorem 5 (central limit theorem).

Let $X_1, X_2, \dots$ be iid random variables with finite mean μ and variance σ^2 . Then

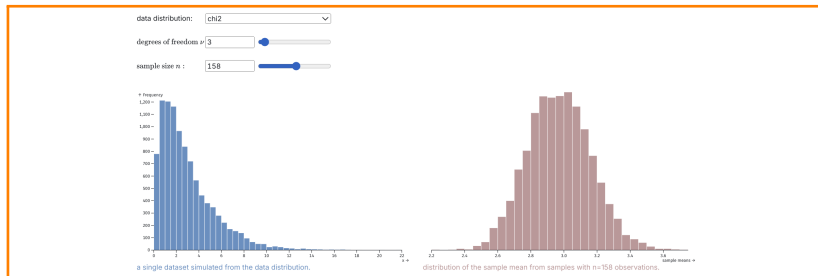
$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1),$$

as $n \rightarrow \infty$, where $\xrightarrow{d}$ denotes convergence in distribution.

Central limit theorem



Central limit theorem - widget



Transformations of random variables

- **Known:** the distribution of X is $f(x)$
- **Wanted:** the **distribution of a transformed variable**

$$Y = g(X)$$

- Why? We often need to **transform the data**.
- Bayes: we often need to transform parameters.
- Examples:
 - ▶ Linear: $Y = a + b \cdot X$
 - ▶ Log: $Y = \log(X)$
 - ▶ Logit: $Y = \log\left(\frac{X}{1-X}\right)$

Transformations of random variables - example

■ Example:

▶ **pdf:** $f_X(x) = 3x^2$ for $0 \leq x \leq 1$

▶ **cdf:** $F_X(x) = \int_0^x 3t^2 dt = [t^3]_0^x = x^3$

■ Linear transformation: $Y = 2 + 3X$

■ cdf of Y :

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) = \Pr(2 + 3X \leq y) = \Pr\left(X \leq \frac{y-2}{3}\right) \\ &= F_X\left(\frac{y-2}{3}\right) = \left(\frac{y-2}{3}\right)^3 \end{aligned}$$

■ pdf of Y

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X\left(\frac{y-2}{3}\right) = f_X\left(\frac{y-2}{3}\right) \cdot \frac{1}{3} \\ &= 3 \left(\frac{y-2}{3}\right)^2 \cdot \frac{1}{3} = \left(\frac{y-2}{3}\right)^2 \quad \text{for } 2 \leq y \leq 5 \end{aligned}$$

Transformations of random variables - example

- A little more general: **linear transformation**: $Y = a + bX$

- **cdf** of Y

$$F_Y(y) = \Pr(Y \leq y) = \Pr(a + bX \leq y) = \Pr\left(X \leq \frac{y-a}{b}\right) = F_X\left(\frac{y-a}{b}\right)$$

- **pdf** of Y

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X\left(\frac{y-a}{b}\right) = f_X\left(\frac{y-a}{b}\right) \cdot \frac{1}{b}$$

- We computed the **inverse transformation**, i.e. solved for x

$$y = a + bx \quad \Longleftrightarrow \quad x = \frac{y-a}{b}$$

- General: if $g(x)$ is an **invertible function**

$$y = g(x) \quad \Longleftrightarrow \quad x = g^{-1}(y)$$

where $g^{-1}(y)$ is the **inverse function**.

Transformations of random variables

■ Transformation formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

■ Need **three piece of information** to apply the formula:

- ▶ The **density** $f_X(x)$ for X
- ▶ The **inverse transformation** $x = g^{-1}(y)$
- ▶ The **derivative** of the **inverse transformation**

$$\frac{d}{dy} g^{-1}(y)$$

■ Note that $|\cdot|$ is the **absolute value** (removes negative signs).

■ For example $|-3| = 3$ and $|5| = 5$.

Transformations of random variables

Transforming variables - change-of-variable formula

Let $X \sim f_X(x)$ and

$$Y = g(X)$$

an invertible monotonically increasing or decreasing transformation with continuous derivative and inverse transformation

$$X = g^{-1}(Y).$$

The density of Y is then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

- If $Y = g(X)$ is piecewise monotone, handle each piece separately and stitch together.
- Example 3 on [Wikipedia on transformations](#) uses this on:
 - ▶ $X \sim N(0, 1)$
 - ▶ $Y = X^2$ which is monotone on $(-\infty, 0)$ and $[0, \infty)$
 - ▶ Result: $Y \sim \chi^2(\nu = 1)$

Transformations of random variables - example

- Let $X \sim N(\mu, \sigma^2)$ with pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

- Let $Y = \exp(X)$ with inverse transformation $X = \log(Y)$ with derivative

$$\frac{d}{dy} g^{-1}(y) = \frac{d}{dy} \log(y) = \frac{1}{y}$$

- Then

$$f_Y(y) = f_X(\log(y)) \cdot \frac{1}{y} = \frac{1}{y\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(\log(y) - \mu)^2\right)$$

for $y > 0$.

- We have shown: if $X \sim N(\mu, \sigma^2)$ then

$$\exp(X) \sim \text{LogNormal}(\mu, \sigma^2)$$

Monte Carlo simulation

- Let $X \sim f(x)$.
- Compute $\mathbb{E}(g(X))$ for some function $Y = g(X)$ by simulation.
- Key idea: **law of large numbers**.
- Simulate $x_1, \dots, x_m \stackrel{\text{iid}}{\sim} f(x)$

$$\frac{1}{m} \sum_{i=1}^m g(x_i) \xrightarrow{p} \mathbb{E}(g(X))$$

- Monte Carlo to **compute tail probability** $\Pr(X > c)$

$$g(x) = \begin{cases} 1 & \text{if } x > c \\ 0 & \text{if } x \leq c \end{cases}$$

$$\begin{aligned} \mathbb{E}(g(X)) &= \int_{-\infty}^{\infty} g(x) f(x) dx = \int_{-\infty}^c 0 \cdot f(x) dx + \int_c^{\infty} 1 \cdot f(x) dx \\ &= \int_c^{\infty} f(x) dx = \Pr(X > c) \end{aligned}$$

Monte Carlo accuracy via the CLT

- Central limit theorem (informal): for large m

$$\frac{1}{m} \sum_{i=1}^m g(x_i) \stackrel{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{m}\right)$$

where

$$\mu = \mathbb{E}(g(X))$$

and

$$\sigma^2 = \mathbb{V}(g(X))$$